

# SolarKits v2.0

## Reseller & EPC Buyer Pipeline Architecture & Commercial Flow Report

### 1. Executive Summary

This document presents the complete architecture, operational workflow, portal integration, and financial revenue engine of the SolarKits v2.0 Reseller Management & EPC Buyer System. The system enables verified Resellers to onboard EPC Buyer sub-accounts in their authorized territories, process solar kit purchases, and generate revenue via dual commercial settlement modes.

### 2. System Access & Portal Mapping

| Portal / Application | Local Dev URL         | Live Production URL                            |
|----------------------|-----------------------|------------------------------------------------|
| Unified Admin Portal | http://localhost:5173 | https://solarkits-unified-admin.onrender.com   |
| Reseller Portal      | http://localhost:5178 | https://solarkits-reseller-portal.onrender.com |
| SolarShop (EPC App)  | http://localhost:5177 | https://solarkits-solarshop-india.onrender.com |
| Central Backend API  | http://localhost:5000 | https://solarkits-central-backend.onrender.com |

#### Verified Test EPC Buyer Credentials:

Portal URL: http://localhost:5177/auth/login | Email: samir@gmail.com | Password: 1234

### 3. End-to-End Workflow Architecture

|        |                                                                                                                                                       |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Step 1 | <b>Reseller Registration &amp; KYC</b><br>Reseller registers on Reseller Portal (Port 5178). Admin verifies KYC and activates account in Admin Panel. |
| Step 2 | <b>Territory Alignment</b><br>Reseller is assigned operating State & District territory rules for authorized pipeline distribution.                   |
| Step 3 | <b>EPC Buyer Onboarding</b><br>Reseller onboards EPC Buyer sub-accounts directly via "My EPC Buyers" pipeline.                                        |
| Step 4 | <b>Admin Review Queue</b><br>Admin reviews and approves EPC Buyer requests in Reseller Management section.                                            |
| Step 5 | <b>EPC Buyer Login &amp; Order</b><br>EPC Buyer logs into SolarShop (Port 5177), sees Reseller Attribution Badge, and places kit orders.              |
| Step 6 | <b>Commercial Settlement</b><br>Order links to Reseller; commission is credited to Reseller Wallet & company secures wholesale profit.                |

## 4. Dual Commercial Settlement & Revenue Engine

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The system features a dual commercial engine ensuring seamless revenue distribution for both Resellers and SolarKits Company:

### Mode 1: Commission Mode (Agency / Affiliate Model)

- EPC Buyer Purchase Price: Standard Retail Price (e.g. ₹1,00,000)
- Reseller Earning: 5% Commission ( ₹5,000) automatically credited to Reseller Wallet Ledger
- SolarKits Company Revenue: Full Payment ( ₹1,00,000) - ₹5,000 Commission - Product Cost = Net Profit
- Reseller Withdrawal: Reseller requests bank transfer payout from Reseller Portal Wallet

### Mode 2: Dealer Mode (Wholesale / Distributor Model)

- EPC Buyer / Dealer Purchase Price: Wholesale Discounted Price (e.g. ₹92,000 - 8% Discount)
- Reseller Earning: ₹8,000 Upfront Margin Discount on invoice purchase
- SolarKits Company Revenue: Direct Cash Payment ( ₹92,000) - Product Base Cost = Guaranteed Profit
- Invoicing: Generates B2B Dealer Tax Invoice with full GST Input Credit

## 5. Director Verification & Demonstration Steps

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1. Open Admin Panel (<http://localhost:5173>) -> Go to SolarShop -> Reseller Management -> EPC Buyers.
2. Verify "samir@gmail.com" is listed with Status: Approved and Attributed Reseller details.
3. Open EPC Buyer Portal (<http://localhost:5177/auth/login>) -> Login using samir@gmail.com / 1234.
4. Observe "Partner Network Badge" in Header and User Profile indicating parent Reseller linkage.
5. Select Solar Combo Kit, proceed to Checkout (<http://localhost:5177/checkout>), and notice Reseller Order Attribution Banner.



